



3D Spatial Deep Learning for Arsenic Risk Prediction

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Abstract

Groundwater arsenic contamination remains a significant public health challenge in Bangladesh, where widespread reliance on tubewells has resulted in millions of people being exposed to unsafe water.

This project investigates the use of a three-dimensional convolutional neural network (3D CNN) for arsenic risk prediction by integrating groundwater measurements, environmental raster data, and spatial features into a voxel-based representation.

As arsenic mobilisation is strongly influenced by groundwater depth and aquifer characteristics, this approach enables the model to learn relationships across both horizontal and vertical dimensions of the subsurface environment.

The results demonstrate the potential of deep spatial learning methods to support scalable arsenic risk assessment.

Table of Contents

Chapter 1	5
Introduction	5
1.1 The Problem	5
1.11 Context	5
1.12 Ongoing Challenges	5
1.2 An Introduction to iArsenic	5
1.21 Overview	5
1.22 Potential for Innovation	5
1.3 Moving Forwards	6
1.31 Research Questions	6
1.32 Rationale	6
1.4 Conclusion	6
Chapter 2	7
Literature Review	7
2.1 Introduction	7
2.2 Pushback	7
2.3 Existing Work	8
2.31 iArsenic	8
2.32 Chemical Predictions using Geospatial Modelling	8
2.33 Conclusion	9
2.4 Python Library Selection	9
2.41 Model Implementation (PyTorch)	9
2.42 Data Formats and Processing (NumPy, Pandas and Rasterio)	9
2.5 Basal Machine Learning Methods	9
2.51 Supervised Learning	10
2.52 Classification vs Regression	10
2.53 Model Evaluation	10
2.6 Summary	11
Chapter 3	12
Reproducible Methodology	12
3.1 Introduction	12
3.2 Requirements	12
3.3 Data Handling	12
3.4 Evaluating a Model	13
3.5 Developing a Model	14

3.6	Conclusion	14
Chapter 4		15
Data Preparation & Applied Methodology		15
4.1	Introduction	15
4.2	Dataset Construction.....	15
4.21	Obtaining and Cleansing Source Data	15
4.22	Spatial Representation	16
4.23	Voxel and Patch Size Selection.....	17
4.24	Hydrogeological Raster Processing.....	18
4.25	Derived Spatial Features.....	19
4.26	Leakage Prevention	20
4.27	Input Channel Overview	20
4.3	Developing the Initial Model	23
4.31	CNN Structure.....	23
4.32	Training Configuration.....	24
4.4	Evaluating Baseline Model.....	24
4.5	Summary	25
Chapter 5		26
Model Development		26
5.1	Introduction	26
5.2	Model 1	26
5.21	Model Overview	26
5.22	Model Performance.....	26
5.23	Identified Limitations	26
5.24	Summary.....	27
5.3	Model 2.....	27
5.31	Model Improvements.....	27
5.32	Summary of Additional Spatial Features	27
5.33	Model Performance.....	28
5.4	Discussion	28
5.5	Chapter Summary.....	28
Chapter 6		29
Evaluation & Further Work		29
6.1	Introduction	29
6.2	Confusion Matrix	29
6.3	Results.....	30
6.4	Future Data Refinement.....	31

6.5	Further Work	32
6.6	Summary	32
Chapter 7		33
Reflection & Conclusion		33
7.1	Proposed Usage of Solution.....	33
7.2	Research Questions Reflection	33
7.21	Research Question 1	33
7.22	Research Question 2	34
7.3	Risks	34
7.4	Project Management	35
7.5	Conclusion	35
References		37
AI Usage		0
Appendices.....		0
Appendix A		0
Project Initiation Document.....		0
Appendix B		1
Ethics Certificate.....		1
Appendix C		3
Code Repository		3

Table of Figures

Figure 1 - Elevation Shapefile - prior to preprocessing	13
Figure 2 - Distribution of wells per mouza across Bangladesh (Histogram)	16
Figure 3 - Elevation Shapefile – preprocessed	18
Figure 4.1 - Example CNN Input for Raster (Safe well)	19
Figure 4.2 - Example CNN Input for Raster (Unsafe well)	19
Figure 5 - Input Channel Overview	29

Table of Tables

Table 1 - Metrics for Model Evaluation.....	10
Table 2 - Distribution of wells per mouza across Bangladesh (Histogram).....	17
Table 3 - Input Channel Overview	20
Table 4 - IArsenic “Model 5” Evaluation	24
Table 5 - Model 1 Performance	26
Table 6 - New Spatial Features	27
Table 7 - Model 2 Performance	28
Table 8 - Combined Metrics Table.....	30
Table 9 - Relative Metrics Table	30

Chapter 1

Introduction

1.1 The Problem

1.11 Context

The contamination of groundwater by arsenic in Bangladesh is the largest poisoning of a population in history, with millions of people exposed. (Smith et al., 2000)

For drinking water, the people of Bangladesh used to rely on surface water, which was often contaminated with life-threatening diseases. To reduce the incidences of these diseases, millions of tubewells were installed. (Frisbie et al., 2002)

The installation of these tubewells led to widespread contamination by arsenic due to vast areas of Bangladesh containing groundwater with concentrations above the World Health Organization (WHO) drinking water standard of 10µg/L. (Smith et al., 1999)

These wells remain widely utilised by citizens of Bangladesh. About 70% of people living in cities and 95% living in rural areas depend on groundwater for domestic and drinking purposes in Bangladesh (Farzana et al., 2025) yet in some villages, 90 percent of the tubewells are unsafe and there are few alternatives to groundwater sources. (Ahmad et al., 2005)

1.12 Ongoing Challenges

Although the mechanisms driving arsenic contamination in Bangladesh are well understood, sustained monitoring of groundwater quality remains a major challenge. Arsenic concentrations vary spatially and temporally due to complex hydrogeochemical processes, meaning that a single test of a tubewell cannot guarantee long-term safety (Smedley & Kinniburgh, 2002; Nickson et al., 2000). Continuous testing is difficult to implement at scale because Bangladesh relies on millions of largely unregulated, privately installed shallow tubewells, many located in remote rural areas (Smith et al., 2000). Laboratory-based arsenic testing is costly, time-consuming, and dependent on trained personnel and infrastructure that are often unavailable outside urban centres (Frisbie et al., 2002). Government and donor-led testing campaigns have therefore been intermittent, with limited capacity for follow-up testing or maintenance of well safety records (Farzana et al., 2025).

1.2 An Introduction to iArsenic

1.21 Overview

iArsenic is a project that encompasses various expert system and machine learning models, featuring a web application for the public to make use of the current leading model.

Built upon decades of groundwater research (iArsenic, n.d.) the approach utilises the latest of its multiple iterations of expert models - algorithms containing rules developed by experts in the field - to predict arsenic contamination of a well using easily measurable well features such as depth, location (Mouza) and staining colour.

1.22 Potential for Innovation

CNNs are widely used for geospatial data analysis due to their ability to automatically learn hierarchical spatial features and capture spatial dependencies within structured data (Wang et al., 2019), despite this, expert systems as described in 1.21 have been opted for in the described project. The origins of arsenic in sediment vary depending on the geology of the area and sediment

components (Hossain M Anawar., 2002) hence one potential criticism towards the iArsenic project is that despite accurately predicting arsenic risk at present using the static features of a well it cannot capture the combined effects of the various environmental factors impacting arsenic mobility. As such, it may be limited in accurately predicting the subsurface transportation of arsenic, what areas are at risk, and why.

Furthermore, while the real-world implications seen by increasing the efficacy of arsenic detection via the use of machine learning have been demonstrated repeatedly throughout these publications, the use of 3D CNNs incorporating depth-dependent groundwater representations for Bangladeshi arsenic risk prediction appears limited within existing publicly available literature.

1.3 Moving Forwards

1.31 Research Questions

1. Is a 3D Convolutional Neural Network (CNN) a more viable approach for arsenic detection?
2. What is the path forward for improving machine learning with chemical detection?

1.32 Rationale

These questions are justified as they serve to benefit the current iArsenic project as well as the people of Bangladesh by considering new approaches to the ongoing issue of groundwater contamination.

Improved predictive accuracy could contribute to more reliable identification of high-risk wells, evaluating both approaches allows for a better understanding of the trade-offs between interpretability, accessibility, and predictive performance, helping determine whether more advanced machine learning techniques provide meaningful practical advantages over existing expert systems.

1.4 Conclusion

This project recognises that it alone cannot eliminate the unprecedented impact of groundwater pollution in Bangladesh to its people. Nevertheless, the following work aims to contribute to the increase in public awareness surrounding the topic and to potentially support evidence-based decision-making, in-turn better directing government and humanitarian aid more effectively toward communities most at risk.

Chapter 2

Literature Review

2.1 Introduction

As stated in 1.12, due to the scale and complexity of the issue the widespread use of laboratory techniques to detect arsenic concentration across the entirety of Bangladesh remains unfeasible, as that remarkable detection power is accompanied by an upfront cost upwards of \$30,000 USD; not to mention the cost of maintenance, consumables, and highly trained technicians. (Yogarajah & Tsai, 2015)

This chapter serves to provide insight into the history of machine learning as an effective alternative method of both arsenic and general chemical detection, aiming to summarise the impact, effectivity and variety in this approach.

2.2 Pushback

The literature surrounding the use of artificial intelligence and machine learning presents a balanced and nuanced perspective regarding both their potential benefits and associated risks.

Scholars emphasise the transformative capabilities of these technologies across multiple sectors. For instance, Gerlich., 2023 argues that all areas of life have the potential to become revolutionised by artificial intelligence. Similarly, Ipsos., 2017 found that participants perceived machine learning in healthcare as one of the areas with the greatest potential to generate benefits for both individuals and society. These perspectives demonstrate the widespread recognition of AI as a valuable tool for improving efficiency, prediction, and decision-making processes.

However, contemporary research also highlights important limitations and ethical concerns that must be carefully considered during implementation. Fleming et al., 2021 note that even highly trained convolutional neural networks may perform unpredictably when exposed to data that differ slightly from their training datasets, illustrating concerns regarding model generalisability and robustness. Furthermore, Tan et al., 2022 argue that machine learning systems inherently reduce a degree of human control when tasks become automated. Concerns regarding transparency are also emphasised by Vorm and Combs., 2022, who advocate for transparency in monitoring, process visibility, surveillance, and disclosure within AI systems.

It is also important to consider the temporal context in which these studies were produced. Much of the literature emphasising the risks and ethical concerns associated with artificial intelligence emerged during the rapid public expansion of generative AI systems, a period characterised by heightened public and academic scrutiny. In contrast, literature highlighting the broader benefits and transformative potential of AI has remained relatively consistent both prior to and following this period. This suggests that, while concerns regarding transparency, reliability, and control have intensified in response to recent developments, the underlying perception of AI as a valuable and impactful technology has remained largely stable.

Nevertheless, none of these critiques found fundamentally reject or discredit the use of machine learning. Rather, they collectively advocate for more responsible, transparent, and carefully justified implementation practices. Consequently, machine learning remains a valid and appropriate methodological approach to complex issues such as arsenic risk detection, provided that developers maintain clear real-world reasoning, transparency, and accountability throughout each stage of the modelling and decision-making process.

2.3 Existing Work

2.31 iArsenic

iArsenic is the most recent and most prominently publicly available project utilising a substantial body of previous peer-reviewed research and relevant data pertaining to arsenic contamination. It features contributions from researchers and experts across multiple countries specialising in closely related fields such as environmental engineering, geology and science. As a result, any potential bias in the work is more likely to arise from imbalanced data distribution in the dataset used to train the model.

The dataset currently featured in the project contains the approximate location, depth (m) and Arsenic concentration (μg) of roughly one million wells, with a new dataset containing data from approximately six million wells to be introduced soon.

The contextual limitations of this dataset may restrict the overall performance of models trained on it. Because arsenic is related to a multisource system, both natural and anthropogenic (Rodriguez et al., 2003), there is great importance in including related information in datasets. Winkel et al., 2008 further highlighted that geological information was of crucial importance in informing their machine learning models. Despite these sources being dated the relevance of these points should not be understated as they still reflect the current state of the field. As such, these shortcomings cannot be ignored.

Although the spatial resolution of the location data is limited to the mouza level (approximately 0.5–5 km²), obtaining more precise geospatial coordinates (latitude and longitude) may be challenging, which in turn would constrain the variability and granularity of the dataset. Consequently, prioritising a larger volume of observations with a richer set of features represents a valid and methodologically justifiable approach. Additionally, as this is one of the few arsenic detection models designed for public-facing use, the adoption of simple input parameters for risk-level estimation is essential and non-negotiable.

2.32 Chemical Predictions using Geospatial Modelling

Barkhordari and Qi (2025) present a recent study investigating the prediction of multiple heavy metal concentrations, including arsenic, across European soils using a multi-end convolutional neural network. Although the study focuses on European soils rather than Bangladeshi groundwater, it provides valuable insight into the integration of geospatial information within deep learning models. As a publication released in 2025, it reflects current developments in environmental machine learning. The paper was received in December 2024, revised in February 2025, accepted later that month, and published online in March 2025, demonstrating a rigorous peer-review process that supports the validity of its findings.

The authors identify three major limitations in existing research. First, they argue that previous studies have been constrained by datasets lacking sufficient resolution and important predictor variables, resulting in the exclusion of factors such as land use intensity, mining activity and infrastructure. This supports the argument that comprehensive spatial datasets are essential for producing accurate environmental prediction models. Similar observations were made by Podgorski et al. (2020), who noted that arsenic measurements were heavily concentrated within a small number of regions and therefore required rebalancing to prevent geographical bias from influencing model performance. Together, these studies emphasise the importance of representative geospatial datasets when developing predictive models.

Secondly, Barkhordari and Qi (2025) note that much of the existing literature focuses on a single contaminant or relies on conventional machine learning methods. Their proposed multi-end CNN demonstrates how deep learning architectures can integrate diverse geospatial variables to model

complex environmental processes more effectively. Finally, the authors highlight that many previously developed models are not publicly accessible, limiting reproducibility and practical adoption. Overall, the study provides objective justification for incorporating diverse geospatial datasets into convolutional neural networks and supports the methodology adopted within this dissertation.

2.33 Conclusion

Existing research demonstrates that accurate arsenic prediction requires the integration of diverse environmental and geospatial factors rather than relying solely on one factor that influences it. Although current datasets such as iArsenic provide valuable large-scale observations, limitations in spatial resolution and available contextual features may restrict predictive performance. Recent advances in geospatial deep learning, particularly the use of convolutional neural networks for environmental modelling, highlight the potential of combining multiple data sources to capture complex relationships. Therefore, this project builds upon previous research by investigating whether a 3D CNN approach can improve arsenic risk prediction through the integration of spatial, geological and chemical information.

2.4 Python Library Selection

This section provides justification for the two primary technical decisions made regarding Python library selection within the project, specifically relating to machine learning model implementation and geospatial image data processing. The chosen libraries were selected based on their suitability for scalability, performance, flexibility, and compatibility with the methodological requirements of the project.

2.41 Model Implementation (PyTorch)

For machine learning model implementation, PyTorch was selected over TensorFlow and scikit-learn. Although TensorFlow remains one of the most widely adopted deep learning frameworks in both academia and industry, PyTorch has become increasingly favoured within contemporary machine learning research due to its intuitive design, dynamic computational graph structure, and greater flexibility during experimentation and debugging. These characteristics are particularly beneficial within research-oriented environments where iterative testing and rapid model modification are required. While scikit-learn provides highly efficient implementations for traditional machine learning methods and lightweight predictive models, it is not optimised for convolutional neural network development or GPU-accelerated deep learning workflows at the scale required for this project. Consequently, PyTorch was determined to be the most suitable framework due to its balance of performance, readability, research adoption, and compatibility with modern 3D CNN architecture.

2.42 Data Formats and Processing (NumPy, Pandas and Rasterio)

For this project, each data processing library was selected based on the dataset types required. Pandas will be used for handling and cleaning tabular data whereas Rasterio will be used for raster-based geological and hydrological data.

NumPy will be used for storing and processing the generated three-dimensional voxel arrays and CNN inputs. Saving these arrays in .npy format will provide efficient loading, reduced storage overhead, and direct compatibility with PyTorch tensors, making it suitable for the repeated multidimensional computations required during 3D CNN training.

2.5 Basal Machine Learning Methods

This section provides justification for the primary methodological decisions made regarding machine learning implementation within the project. Each decision is accompanied by practical real-world reasoning to ensure that the proposed approach remains both technically effective and appropriate with consideration towards public-facing arsenic risk prediction.

2.51 Supervised Learning

Within the context of supervised learning, V. Nateski (2017) describes machine learning as a process of finding statistical regularities or other patterns in data.

A supervised learning approach was selected because the available dataset is both labelled and structured, with each well associated with a known arsenic concentration. This enables the model to learn relationships between environmental variables and arsenic contamination by identifying patterns within the training data.

D. Valkenborg (2023) argues that one challenge of supervised learning is determining whether the examples are representative enough to cover the learning objective. However, this limitation is mitigated by the large-scale iArsenic dataset, comprising over 740,000 usable well observations, together with multiple hydrogeological raster datasets that provide comprehensive spatial coverage across Bangladesh. As a result, the training data are considered sufficiently representative for the prediction task undertaken in this dissertation.

2.52 Classification vs Regression

The project incorporates both regression and classification approaches, mirroring the methodology currently utilised by the iArsenic project. Regression outputs will be used to estimate a specific arsenic concentration value, while classification outputs will categorise wells into two predefined risk levels. However, greater emphasis is placed upon classification, as this project prioritises accessibility and usability for end users. Specific arsenic concentration values may hold little meaning for members of the public without technical or medical context, whereas clearly defined risk categories provide more understandable and actionable information.

Rather than using arbitrary thresholds, this project defines arsenic risk as a binary classification problem based on established drinking water safety standards. Wells containing arsenic concentrations less than or equal to 10 µg/L will be classified as safe, reflecting the World Health Organization guideline for acceptable arsenic levels in drinking water. Wells exceeding this threshold will be classified as unsafe due to the increased health risks associated with long-term arsenic exposure.

This binary classification approach was selected to provide a clear distinction between compliant and non-compliant water sources, improving interpretability for practical risk assessment. Additionally, reducing the problem to two classes avoids ambiguity between intermediate contamination levels and allows the model to focus on the key objective of identifying wells that exceed safe drinking water limits.

2.53 Model Evaluation

Model performance and comparison will be evaluated using metrics in the table below.

Table 1
Metrics for Model Evaluation

<i>True Positive</i>	<i>True Negative</i>	<i>f1</i>	<i>Accuracy</i>	<i>Precision</i>
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These metrics were selected because they were also utilised within the original iArsenic project, allowing for direct comparison between approaches. Furthermore, this document will showcase confusion matrices as they will provide valuable insight into the types of classification errors being made which is particularly important in public health contexts where false negatives may have severe real-world consequences.

2.6 Summary

This chapter reviewed existing machine learning approaches for arsenic prediction and highlighted the importance of incorporating comprehensive datasets and geospatial information. Limitations within current methods, including spatial resolution and feature availability, provide justification for the proposed 3D CNN approach. The selected machine learning methodology, evaluation metrics, and technical frameworks have been discussed to support future decisions and ensure accurate, interpretable, and practical arsenic risk prediction.

Chapter 3

Reproducible Methodology

3.1 Introduction

This chapter outlines the systematic process used to develop, evaluate, and refine the proposed model to ensure transparency and reproducibility. An iterative development approach is employed to progressively improve both the quality of the dataset and 3D CNN architecture through the evaluation of existing models, literature, and repeated experimentation.

The methodology encompasses requirement analysis, research, dataset production, model development, and performance evaluation. Collectively, these stages will provide a structured framework for producing a reliable, scalable, and reproducible depth-based machine learning model suitable for public-facing arsenic risk prediction.

3.2 Requirements

The methodology begins by defining requirements derived from the research objectives and decisions established in previous chapters. These requirements provide a framework for developing and evaluating the proposed 3D CNN arsenic risk prediction model.

The system aims to provide an accessible method for estimating groundwater arsenic risk without requiring laboratory testing. Therefore, the model must prioritise reliable and interpretable predictions while remaining suitable for practical use in areas where water quality monitoring is limited.

Technical requirements include the use of appropriate preprocessing, feature selection, and training strategies to reduce bias and improve generalisation. All changes to the dataset, model architecture, and training process will be justified through experimental evaluation or existing research to ensure a transparent and reproducible methodology.

The model must also consider ethical limitations, with predictions being presented as risk estimates rather than replacements for laboratory measurements. Evaluation will focus on improving predictive performance, particularly reducing false negative classifications, as incorrectly identifying contaminated wells as safe presents the greatest public health risk.

3.3 Data Handling

The performance of a convolutional neural network is highly dependent on the quality and consistency of the data used during training. Therefore, all datasets incorporated within this project will be obtained from authoritative sources and, where possible, collected over similar time periods to minimise temporal bias. Combining datasets captured years apart may introduce inconsistencies resulting from environmental change, differences in collection methods, or varying spatial resolutions. Consequently, preference will be given to data published by recognized and preferably government organisations and websites (e.g. Bangladesh Water Development Board).

Prior to preprocessing, all geospatial datasets will be inspected and processed using QGIS, an open-source Geographic Information System widely used for analysing spatial data. QGIS enables raster layers, such as Digital Elevation Models (DEMs), to be visualised and verified for coordinate reference systems, spatial resolution, and geographical alignment before being incorporated into the machine learning pipeline. Datasets must be clipped to the study area and transformed into a common coordinate reference system as not doing so can cause value-location mismatches or out-

of-bounds errors. It is also preferable to transform values into a consistent scale, as machine learning models can be influenced by features with larger numerical ranges, causing them to dominate the learning process and reducing the contribution of smaller-scale variables.

Figure 1



Elevation Shapefile - prior to preprocessing

Note. Screenshot created using QGIS contour line data from data.humdata.org/dataset/bangladesh-contour-lines.

3.4 Evaluating a Model

Model evaluation will be used to objectively assess the impact of changes made during development and ensure that improvements are measurable and reproducible. Each model iteration will be

evaluated before further modifications are introduced, allowing ineffective changes to be identified and successful approaches to be retained.

Performance will be assessed using accuracy, precision, F1 score, true positive rate, and true negative rate to enable comparison with existing iArsenic models. While accuracy provides an overall measure of performance, greater emphasis will be placed on precision and F1 score due to their consideration of classification errors. Reducing false negatives will be a key priority, as incorrectly identifying unsafe wells as safe presents the greatest public health risk.

Confusion matrices will also be analysed to identify specific classification patterns and errors, while training and validation loss curves will be monitored to assess model convergence and detect overfitting. Together, these measures will provide a comprehensive evaluation of the proposed 3D CNN approach.

3.5 Developing a Model

Several factors will influence the design of the CNN architecture. Model complexity must remain appropriate for the size and variability of the available dataset. Increasing the number of trainable parameters allows the network to learn more complex spatial relationships from the 3D voxel representations; however, excessive complexity can increase computational requirements and the risk of overfitting. Conversely, an overly simple architecture may fail to capture important relationships between environmental, spatial, and depth-dependent features.

The balance between network depth and width can therefore be evaluated experimentally. Additional convolutional layers can enable the extraction of higher-level spatial patterns, while increased filter numbers allow the model to learn a wider range of features. Architectural decisions, including kernel size, pooling strategy, activation functions, optimisation method, learning rate, and regularisation techniques, will be assessed based on their impact on validation performance and generalisation.

Each modification to the architecture will be evaluated independently to identify the most effective configuration for predicting groundwater arsenic risk using three-dimensional spatial information.

3.6 Conclusion

This chapter established the methodology for developing the proposed 3D CNN arsenic risk prediction model. The key decisions regarding data processing, model development, architectural design, and evaluation were defined to ensure a structured and reproducible approach. These foundations guide the implementation and experimentation presented in the following chapter.

Chapter 4

Data Preparation & Applied Methodology

4.1 Introduction

Chapter 3 introduced the theoretical concepts, machine learning techniques, and geospatial methodologies that underpin this research. Building upon this foundation, this chapter describes how these concepts were applied to develop a three-dimensional convolutional neural network (3D CNN) for groundwater arsenic risk prediction in Bangladesh. The chapter details the implementation of the proposed methodology, including dataset construction, spatial representation through voxelisation, hydrogeological raster processing, derived feature generation, CNN architecture, and the baseline model used for evaluation. Together, these sections demonstrate how the theoretical framework established in the previous chapter was translated into a practical deep learning system.

4.2 Dataset Construction

The dataset construction process involved transforming raw well measurements and spatial datasets into a structured format suitable for training the 3D CNN. This included cleaning and filtering the well data, generating spatial representations through voxelisation, and preprocessing environmental raster layers. QGIS was used throughout this process to visualise distributions, validate spatial alignment, and prepare geospatial inputs by clipping and transforming datasets to the Bangladesh study area.

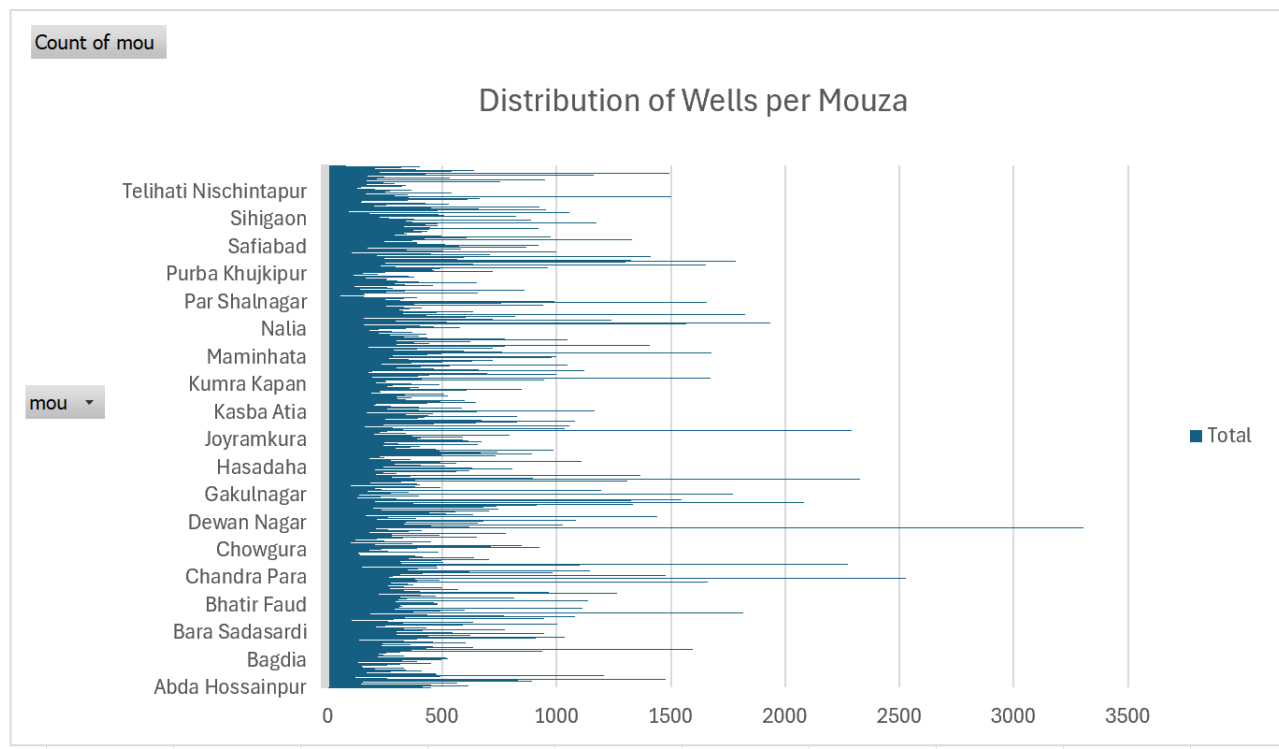
4.21 Obtaining and Cleansing Source Data

The primary well dataset used in this study was obtained from the publicly available iArsenic GitHub repository, which compiles arsenic measurements from multiple reputable studies and data sources. The dataset contains information for 955,892 wells across Bangladesh, including arsenic concentration, well depth, and location recorded at the administration 5 (mouza) level.

To enable spatial modelling, mouza-level locations were converted into geographic coordinates using available administrative boundary data. The well locations are now represented using mouza centroids, where each well inherited the geographic position of its corresponding administrative region. These locations must be transformed into a consistent coordinate reference system before voxelisation to ensure accurate spatial measurements and alignment with environmental raster datasets. To simplify subsequent spatial processing and ensure all distance measurements are represented in consistent metric units, all spatial datasets were transformed into a common projected coordinate reference system (CRS) using EPSG:32646. Using a single projected CRS ensures that distances, voxel dimensions, and raster alignment are measured consistently throughout the dataset preparation process.

Wells with missing coordinates or incomplete measurements are to be removed prior to training, this results in a final dataset of 741,741 usable well observations. These wells provide the foundation for generating the spatial inputs required by the proposed 3D CNN model.

To improve the reliability of local spatial statistics, mouzas containing fewer than seven wells were excluded. This threshold was selected to reduce noise from sparsely sampled regions while maintaining sufficient geographic coverage across Bangladesh. The remaining wells (740,743 - 99.87% of the previously usable wells) were then used to calculate derived spatial features, such as local arsenic distributions and neighbouring well statistics, which were incorporated into the CNN input representation.

Figure 2*Distribution of wells per mouza across Bangladesh (Histogram)*

Note. This is an overview and the y-axis labels do not show every mouza present in the dataset.

4.22 Spatial Representation

Following preprocessing, the well observations are converted into a three-dimensional voxel representation to allow the CNN to learn spatial relationships between arsenic measurements, geological conditions, and environmental variables. By converting the dataset into a three-dimensional representation, neighbouring wells, depth information, and environmental characteristics can be analysed together at each layer, allowing the CNN to identify spatial patterns associated with arsenic risk, which depth is heavily correlated with.

The voxelisation process divided the study area into fixed horizontal and vertical intervals, with each voxel representing a 250m by 250m by 10m volume. A horizontal resolution of 250m was selected as a balance between spatial detail, computational requirements, and data density. Smaller voxel sizes would provide increased spatial resolution but would significantly increase sparsity, resulting in many voxels containing few or no observations. This would reduce the reliability of calculated local statistics and increase computational complexity. Conversely, larger voxels would reduce spatial precision and risk combining areas with different geological and hydrogeological characteristics. A 250m resolution therefore provides sufficient detail to represent local variation while maintaining meaningful well coverage.

A vertical resolution of 10m was selected to preserve depth-related information, as arsenic contamination varies significantly between shallow and deeper groundwater systems. A smaller

depth interval would increase the number of empty voxels and computational cost, while a larger interval would reduce the model's ability to distinguish between different aquifer depths. A 10m depth resolution provides a compromise between capturing vertical variation and maintaining a practical number of input features. This is particularly important in Bangladesh, where shallow and deeper tube wells can exhibit substantially different arsenic risk profiles.

Using the previously mentioned constraints during voxelisation, 33,572 voxels contained valid data and were therefore classified as occupied voxels. These occupied voxels were used to generate local spatial features, including arsenic distributions, neighbouring well statistics, and hydrogeological information. Empty voxels were retained within the three-dimensional structure because their absence still provides important spatial context. These regions were informed through neighbouring occupied voxels and environmental raster layers, allowing the CNN to learn patterns beyond only directly sampled locations and more closely mimic the spread of arsenic.

4.23 Voxel and Patch Size Selection

Table 2
Region Sizes and Mean Data

Administrative Level	Mean Area km ²	Mean Data	Unique Values
Division	17,482	108,584	8
District	2,185	13,778	63
Upazila	257	1,952	445
Union	27	306	2,838
Mouza	2.4	90	9,550

For each target well, a three-dimensional input patch was generated surrounding its corresponding voxel. The selected patch size was 2.25km by 2.25km by 50m, equivalent to a $9 \times 9 \times 5$ voxel region (405 voxels total). This size was chosen to provide sufficient local context while maintaining computational efficiency. The horizontal extent was selected because arsenic contamination exhibits spatial continuity over kilometre-scale distances, meaning nearby wells and environmental conditions can provide useful information for estimating an unsampled location. A 2.25km horizontal area may also provide coverage beyond the target mouza, allowing the model to incorporate information from neighbouring administrative regions rather than relying solely on a single area.

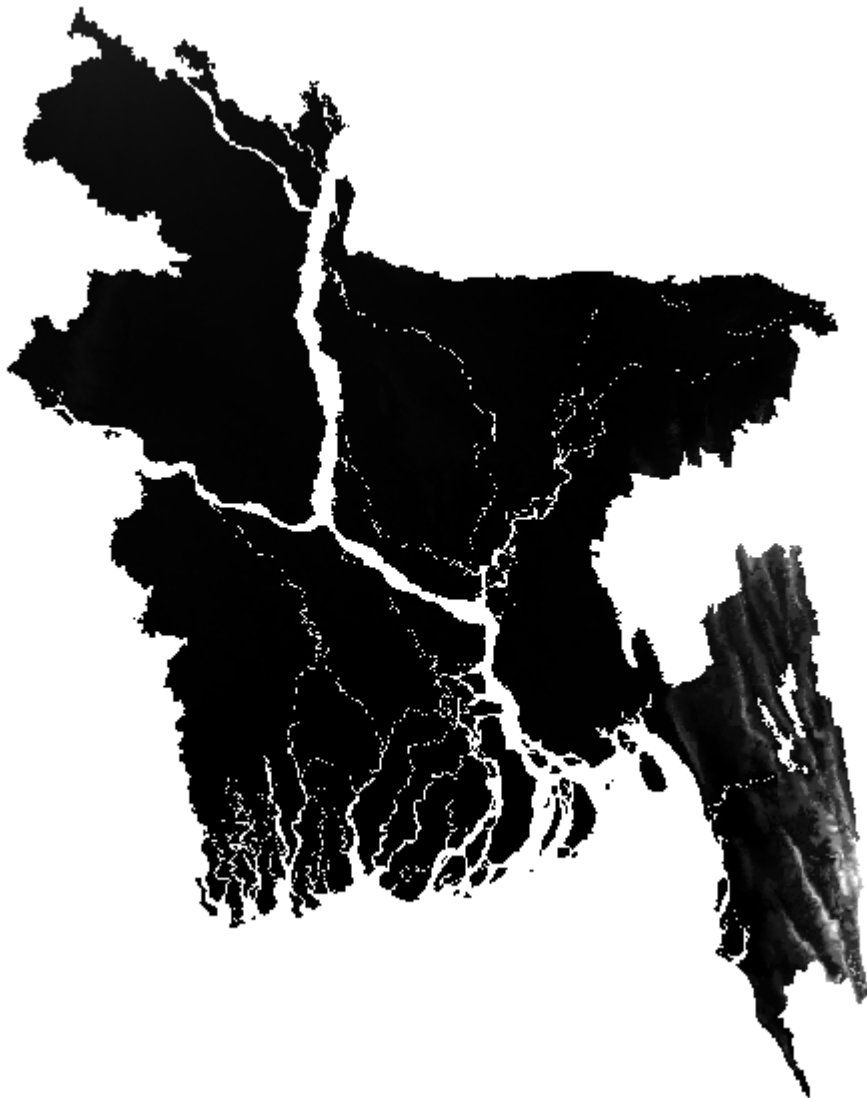
The vertical patch depth of 50m was selected to capture relevant groundwater variation. Arsenic risk is strongly related to well depth and the model requires sufficient vertical context to distinguish between shallow and deeper groundwater conditions. Using five 10m voxel layers will provide enough depth information for the CNN to learn vertical patterns without introducing excessive computational overhead.

These parameters were selected based on the spatial characteristics of the dataset, existing knowledge of arsenic distribution, and computational constraints, while remaining adjustable if required during model development to evaluate their impact on prediction performance.

4.24 Hydrogeological Raster Processing

Several hydrogeological raster datasets were incorporated to provide environmental context alongside the well observations. These datasets are obtained from publicly available and reputable sources and processed within QGIS before being used by the model. The raster layers consisted of elevation, geology, flood risk category and river distance, each providing complementary information about the physical environment that may influence groundwater arsenic distribution.

Figure 3

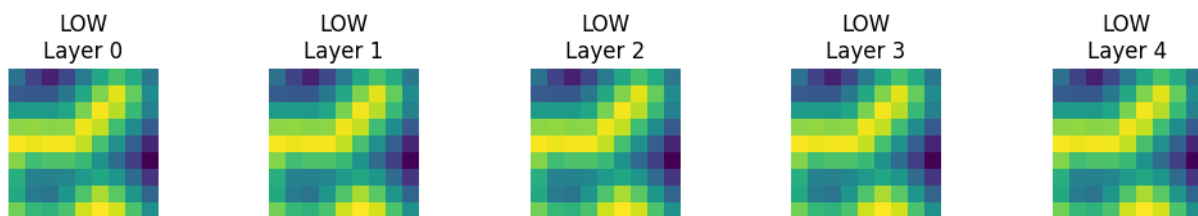


Elevation Shapefile - preprocessed

Note. The elevation map appears largely as a uniform colour because Bangladesh has relatively little topographic variation. Consequently, the elevation range is narrow, causing QGIS to render the raster with minimal visible contrast.

Prior to model training, all raster datasets were clipped to the Bangladesh study area, transformed into a common projected coordinate reference system, resampled to a consistent spatial resolution (250m), and aligned to a common grid to ensure spatial consistency. Continuous raster layers were normalised using z-score standardisation, while embeddings were used on categorical datasets to preserve discrete class information. The preprocessing pipeline was designed to be modular, allowing raster layers to be easily added or removed to evaluate their contribution to model performance. The individual datasets and their justification are summarised in Table X.

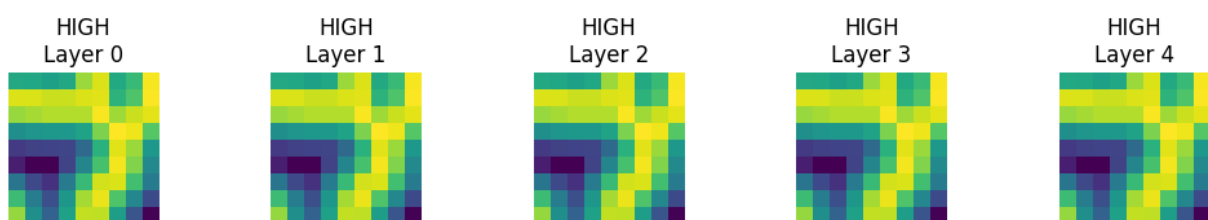
Figure 4.1



Example CNN Input for Raster (Safe well)

Note. Figure 4.1 illustrates example model inputs for a low-risk well at each 10m depth slice sourced from the river distance raster.

Figure 4.2



Example CNN Input for Raster (Unsafe well)

Note. Figure 4.2 illustrates example model inputs for a high-risk well at each 10m depth slice sourced from the river distance raster.

4.25 Derived Spatial Features

To enhance the information available to the CNN by reducing the quantity of empty voxels, additional spatial features were derived from the voxelised well data. These features were generated using neighbouring wells and occupied voxels to represent local arsenic distributions and spatial

confidence. Inverse Distance Weighting (IDW) was applied to estimate arsenic characteristics in unsampled regions, while voxel statistics such as mean concentration, percentiles, and risk distributions captured local variation. Confidence features were also included to indicate the reliability of spatial estimates based on well density and distance to observations, reducing any potential negative impact of sparse sampling areas.

4.26 Leakage Prevention

To prevent the CNN from learning directly from the target well, spatial features were generated while excluding the observation being predicted. As the objective is to predict a new well using nearby existing observations, derived features require careful handling.

Voxel statistics and IDW features were dynamically calculated during data generation, with the target well removed from local calculations. This ensures that features such as arsenic distributions, percentiles, and risk fractions represent surrounding spatial information rather than the target value itself.

4.27 Input Channel Overview

Table 3
Input Channel Overview

<i>Channel</i>	<i>Type</i>	<i>Source</i>	<i>Purpose</i>
1 - Distance to River	Raster	HydroRivers Version 1 (www.hydrosheds.org)	Represents influence of river systems on groundwater flow, sediment deposition, and arsenic transport
2 - Elevation		Bangladesh – Contour Lines (data.humdata.org)	Captures terrain variation affecting groundwater recharge, flow pathways, and arsenic distribution
3 to 34 - Geology		Surface Geology of Bangladesh (www.sciencebase.gov)	Represents subsurface mineral composition and sediment characteristics influencing arsenic mobilisation
35 to 42 - Flood Category		Flooding Levels & Vulnerability (data.humdata.org)	Captures flood-related sediment processes and groundwater conditions associated with arsenic variation
43 - Well Count	Value	iArsenic Github	Indicates local sampling density and reliability of

		spatial arsenic estimates
44 - Arsenic Mean	Calculated Statistic	Provides average nearby arsenic concentration to represent local contamination trends
45 - Arsenic Median	Calculated Statistic	Provides robust local arsenic concentration estimates while reducing influence from extreme values
46 - Arsenic p10	Calculated Statistic	Represents lower-end local arsenic distribution
47 - Arsenic p90	Calculated Statistic	Represents higher-end local arsenic distribution and contamination potential
48 - Arsenic p25	Calculated Statistic	Captures lower quartile variation in local arsenic concentrations
49 - Arsenic p75	Calculated Statistic	Captures upper quartile variation in local arsenic concentrations
50 - Arsenic p95	Calculated Statistic	Represents extreme arsenic concentrations and potential hotspots
51 - IQR	Calculated Statistic	Measures local arsenic variability and spatial heterogeneity
52 - p95 minus p90	Calculated Statistic	Captures variation among high arsenic concentrations
53 - p25 minus p10	Calculated Statistic	Captures variation among low arsenic concentrations
54 - Depth Mean	Value	Represents typical groundwater depth, influencing arsenic

			occurrence through aquifer conditions
55 - Depth Std	Value		Represents variation in sampling depth and differences between groundwater layers
56 - Confidence	Calculated Statistic		Indicates reliability of local arsenic interpolation based on available observations
57 - Relative X	Value		Provides horizontal spatial context for neighbouring arsenic patterns
58 - Relative Y	Value		Provides horizontal spatial context for neighbouring arsenic patterns
59 - Absolute Depth	Value		Captures depth-dependent arsenic variation within aquifer layers
60 - Depth Difference	Value		Represents vertical separation between wells, influencing similarity of groundwater conditions
61 - Chem FE	Value	DPHE/BGS National Hydrochemical Survey	Iron minerals can contribute to arsenic mobilisation through adsorption and release processes
62 - Chem Mn	Value		Manganese cycling influences arsenic release under changing groundwater conditions
63 - Chem SO4	Value		Sulphate availability can influence redox conditions affecting arsenic mobilisation

64 - Chem Ca	Value	Represents groundwater chemistry conditions associated with arsenic behaviour
65 - Chem Mg	Value	Represents mineral dissolution and groundwater chemistry variation
66 - Chem Na	Value	Indicates groundwater salinity and geochemical conditions related to arsenic occurrence
67 - Chem Si	Value	Represents silicate mineral interactions affecting groundwater chemistry
68 - Chem P	Value	Represents chemical conditions that may influence arsenic mobility
69 - Chem Distance	Value	Indicates confidence of chemical estimates based on proximity to sampled locations
70 - Chem Count	Value	Represents availability of nearby groundwater chemistry measurements

Note. Values denote variables directly available within the original dataset; Calculated statistics denote features derived through combinations or further processing of existing variables; Raster denotes variables extracted from geospatial raster (.tif) files.

4.3 Developing the Initial Model

4.31 CNN Structure

The proposed 3D CNN architecture will extract spatial features from volumetric groundwater representations. The model will use raster-based environmental features combined with spatial and statistical channels as input, allowing it to learn relationships across horizontal and vertical dimensions.

The network will consist of four 3D convolutional layers, increasing feature depth from 24 to 192 channels. Each convolutional layer will use 3 x 3 x 3 voxel kernels with padding, followed by Group Normalisation and ReLU activation to improve training stability. Two max-pooling layers will reduce horizontal resolution while preserving the depth dimension, allowing the model to retain depth-dependent arsenic patterns within aquifer layers.

Following feature extraction, adaptive average pooling will compress the horizontal dimensions while maintaining five depth representations. The resulting features will be flattened and passed through a fully connected layer to generate a 256-dimensional embedding, with dropout applied to reduce overfitting. This learned representation will then be used for arsenic risk classification and concentration prediction.

The architecture will enable the model to integrate environmental, spatial, and subsurface information to identify patterns associated with groundwater arsenic contamination.

4.32 Training Configuration

The dataset will be divided using an 80/20 random split, with 80% of samples allocated for training and 20% reserved for validation.

A random split was selected because the objective of this study is to predict arsenic risk at unsampled wells using surrounding groundwater observations and environmental information. Since the model is designed to utilise local spatial relationships through IDW features and voxel-based representations, nearby wells provide valuable contextual information rather than representing independent geographic regions.

4.4 Evaluating Baseline Model

Table 4
iArsenic “Model 5” Evaluation

<i>True Positive</i>	<i>True Negative</i>	<i>f1</i>	<i>Accuracy</i>	<i>Precision</i>
0.814367	0.854302	0.818085	0.836246	0.821837

The proposed 3D CNN approach will be evaluated against the existing iArsenic model, which is available through GitHub (github.com/portsoc/iArsenic) which will be used as the benchmark for this study. The benchmark model “Model 5” provides a reference point for determining whether three-dimensional deep learning approaches can achieve competitive performance or provide additional predictive capabilities for arsenic risk modelling. The evaluation will use the same metrics defined in Chapter 3 to ensure a consistent comparison.

As shown in Table X, the iArsenic benchmark establishes an impressive baseline performance required for assessing the effectiveness of the proposed 3D CNN model.

The objective of this comparison is not solely to exceed the benchmark accuracy, but to determine whether 3D CNNs can capture meaningful spatial relationships within groundwater data. By representing the environment as a three-dimensional volume, the proposed approach aims to learn relationships between depth, surrounding wells, and environmental variables that may not be fully represented by this existing method. Therefore, improvements over the benchmark would

demonstrate the value of volumetric feature learning, while comparable performance would still indicate potential benefits through improved spatial representation and interpretability.

4.5 Summary

This chapter described the specific methodology used to develop this arsenic risk prediction model, including data processing, feature engineering, voxel generation, model architecture, and evaluation considerations. Leakage prevention and benchmark selection were also discussed to ensure reliable comparison. This provides the foundation for evaluating whether 3D CNNs can improve arsenic risk prediction through enhanced spatial feature representation.

Chapter 5

Model Development

5.1 Introduction

Chapter 4 presented the design and implementation of the proposed 3D CNN for arsenic risk prediction. This chapter describes the iterative development of the model, with each version designed to address limitations identified in the previous iteration. To ensure fair comparison, all models were trained and evaluated using the same dataset and methodology described in Chapter 4, with improvements resulting solely from changes to feature engineering or model architecture. The impact of each modification is evaluated to determine its contribution to overall predictive performance.

5.2 Model 1

5.21 Model Overview

Model 1 represents the first implemented version of the proposed 3D CNN architecture. The objective of this model was to establish a functional baseline capable of predicting arsenic risk using the developed voxel-based representation. The model incorporated environmental raster inputs, derived spatial statistics, groundwater depth information, and local arsenic distribution features to provide both environmental and subsurface context to the network.

The purpose of this iteration was not to achieve final predictive performance, but to evaluate the feasibility of the proposed approach and identify limitations within the initial implementation. These findings were used to guide further development and inform the modifications introduced in subsequent models.

5.22 Model Performance

Table 5
Model 1 Performance

<i>True Positive</i>	<i>True Negative</i>	<i>f1</i>	<i>Accuracy</i>	<i>Precision</i>
0.946	0.935	0.937	0.940	0.927

The results demonstrate that the initial model was capable of learning meaningful spatial patterns from the voxel representation, already surpassing the IArsenic “Model 5” metrics.

However, analysis of the model performance identified limitations in the representation of local arsenic distributions and the preservation of subsurface depth information. These limitations informed the feature and architectural modifications introduced in Model 2.

5.23 Identified Limitations

Analysis of Model 1 identified two main areas for improvement. Firstly, the model may have been losing some vertical spatial information during feature extraction. As the CNN processes the three-dimensional voxel representation, pooling operations reduce spatial dimensions to create increasingly abstract features. While this improves computational efficiency, it may also reduce the

network's ability to distinguish relationships between arsenic concentrations and groundwater depth, which is an important factor in subsurface contamination patterns.

Secondly, the model had limited information describing the local distribution of arsenic risk classification within surrounding wells. Although existing spatial statistics provided general information about nearby arsenic concentrations, additional features could be inferred from neighbouring wells to better represent the risk distribution of the surrounding area. Incorporating features describing the proportion and proximity of different risk categories could provide the model with a more complete representation of local hydrogeological conditions without introducing the target well information.

5.24 Summary

Model 1 demonstrated the feasibility of using a voxel-based 3D CNN approach for arsenic risk prediction. The initial results exceeded the benchmark, showing that the proposed spatial representation could capture meaningful relationships between environmental conditions and arsenic risk.

However, analysis of the model identified areas where additional information and architectural improvements could further enhance performance. These findings motivated the development of Model 2, which introduced additional spatial distribution features and modifications to improve the preservation of subsurface depth information.

5.3 Model 2

5.31 Model Improvements

Model 2 introduced two major modifications based on the limitations identified in Model 1: improved preservation of depth information and additional spatial distribution features.

The first modification altered the pooling strategy used within the CNN. Instead of reducing all three spatial dimensions simultaneously, pooling was applied only across the horizontal dimensions while retaining the vertical depth dimension. This allowed later layers of the network to maintain access to depth-dependent information, improving the model's ability to learn relationships between groundwater depth and arsenic risk.

The second modification introduced six additional channels describing the distribution of surrounding wells. These features represented the proportion and proximity of neighbouring wells across different arsenic risk categories, providing the model with additional information about local contamination patterns. To prevent data leakage, the target well was excluded from all calculations, with statistics dynamically generated using only surrounding observations. This ensured that the additional features represented the surrounding hydrogeological environment rather than directly revealing the prediction target.

5.32 Summary of Additional Spatial Features

Table 6
New Spatial Features

<i>Channel</i>	<i>Type</i>	<i>Source</i>	<i>Purpose</i>
71 - Low Risk %	Calculated Statistic	IArsenic Github	Represent well arsenic distribution of an area

72 - Medium Risk %	Calculated Statistic	Represent well arsenic distribution of an area
73 - High Risk %	Calculated Statistic	Represent well arsenic distribution of an area
74 – Distance to nearest low risk well	Calculated Statistic	Represent well arsenic distribution of an area
75 – Distance to nearest medium risk well	Calculated Statistic	Represent well arsenic distribution of an area
76 – Distance to nearest high risk well	Calculated Statistic	Represent well arsenic distribution of an area

5.33 Model Performance

Model 2 was evaluated metrics described in previous chapters. The final results of this project are presented below, which will be discussed in further detail in the following chapter.

Table 7
Model 2 Performance

<i>True Positive</i>	<i>True Negative</i>	<i>f1</i>	<i>Accuracy</i>	<i>Precision</i>
0.972	0.951	0.959	0.962	0.968

5.34 Discussion

The modifications introduced in Model 2 achieved their aim to improve the model's representation of the spatial environment. Preserving the depth dimension allowed the CNN to retain more information about vertical groundwater structure, while the additional channels provided a more detailed representation of surrounding well distributions.

5.5 Chapter Summary

This chapter described the iterative refinement of the proposed 3D CNN architecture. Model 1 established the initial feasibility of the voxel-based approach and identified limitations within the original representation. Model 2 addressed these limitations through modifications to the pooling strategy and the introduction of additional spatial features. The performance of the developed models is analysed further in Chapter 6.

Chapter 6

Evaluation & Further Work

6.1 Introduction

This chapter evaluates the final developed model against the original benchmark and previous iterations. The evaluation follows the methodology defined in Chapter 3, using the same performance metrics to ensure consistent comparison. The results are analysed through classification metrics, confusion matrix analysis, and comparison against the iArsenic benchmark to determine the effectiveness of the proposed 3D CNN approach.

6.2 Confusion Matrix

The confusion matrix provides further insight into the classification behaviour of the final model beyond the aggregated evaluation metrics.

Figure 5

		Predicted	
		Positive	Negative
Actual	Positive	0.97	0.03
	Negative	0.05	0.95

Model 2 Confusion Matrix

The results demonstrate that the model correctly identified 97.2% of unsafe wells and 95.1% of safe wells within the validation dataset. This indicates that the model was effective at distinguishing between groundwater sources that exceeded the defined arsenic safety threshold and those that remained within acceptable limits.

Within the context of arsenic risk prediction, false negative classifications represent the most significant concern because they indicate cases where contaminated groundwater may be incorrectly considered safe. The final model produced a false negative rate of 2.8%, meaning that only a small proportion of unsafe wells were incorrectly classified. This represents a desirable outcome for a public health-oriented prediction system, where prioritising the identification of potentially unsafe sources is particularly important.

False positive classifications are also relevant because incorrectly identifying safe wells as contaminated may lead to unnecessary testing or concern. The model produced a false positive rate of 4.9%, indicating that while some safe wells were incorrectly classified as unsafe, this occurred less frequently than correct classifications. The balance between these errors demonstrates that the model was able to maintain strong sensitivity towards unsafe wells without excessively increasing false alarms.

However, these results should be interpreted within the limitations of the available validation approach. As discussed in Chapter 4, the current evaluation uses a random train-validation split, meaning that neighbouring spatial regions may be represented across both training and validation data. While this approach is suitable for evaluating prediction at unsampled wells within the same geographical distribution, additional testing using geographically separated validation data would provide further evidence of model generalisation.

6.3 Results

The final model has been evaluated against the iArsenic benchmark. The comparison demonstrates the relative effectiveness of the proposed 3D CNN approach.

Table 8
Combined Metrics Table

<i>Model</i>	<i>True Positive</i>	<i>True Negative</i>	<i>f1</i>	<i>Accuracy</i>	<i>Precision</i>
iArsenic "Model 5"	0.814	0.854	0.818	0.836	0.822
Baseline Model (Model 1)	0.946	0.935	0.937	0.940	0.927
Final Model (Model 2)	0.972	0.951	0.959	0.962	0.968

Compared with the iArsenic benchmark, the final model improved across all evaluated metrics. Accuracy increased from 0.836 to 0.962, while the F1 score increased from 0.818 to 0.959. Improvements were also observed in both true positive and true negative rates, indicating that the model was able to improve identification of both contaminated and uncontaminated wells.

Table 9
Comparative Metrics Table

<i>Model</i>	<i>True Positive</i>	<i>True Negative</i>	<i>f1</i>	<i>Accuracy</i>	<i>Precision</i>
iArsenic "Model 5"	0	0	0	0	0

Baseline Model (Model 1)	+0.132	+0.081	+0.119	+0.104	+0.105
Final Model (Model 2)	+0.158	+0.097	+0.141	+0.126	+0.146

The improvement over the benchmark suggests that incorporating environmental and spatial information into a three-dimensional representation provides additional predictive capability compared with approaches based primarily on individual well characteristics. Rather than treating each well as an independent observation, the proposed approach allows surrounding groundwater conditions, environmental characteristics, and depth information to contribute towards risk estimation.

Although the improvement demonstrates the potential of three-dimensional deep learning for arsenic prediction, performance metrics alone do not fully represent practical suitability. A model designed for public health applications must also consider reliability, interpretability, and limitations in the underlying data. The predictions produced by this model should therefore be considered risk estimates that can support decision-making rather than replacements for direct laboratory testing.

The results indicate that the proposed methodology successfully achieved its primary objective of investigating whether CNN-based spatial learning could provide improvements over existing arsenic prediction approaches. The findings support the hypothesis that representing groundwater conditions as a volumetric structure enables a model to capture relationships that may not be available through simpler feature-based approaches.

6.4 Future Data Refinement

Although the dataset used within this project provides extensive coverage across Bangladesh, further improvements may be possible through the availability of additional and higher-quality data sources.

One potential improvement would be obtaining more accurate well coordinates. The current dataset relies on mouza-level spatial information, meaning individual wells are represented using the centroid location of their administrative area. While this provides extensive geographic coverage, it introduces spatial uncertainty because multiple wells within the same mouza may be positioned at the same location despite being distributed across the wider region. More precise well coordinates would allow the voxel representation to better capture local variation and improve the accuracy of neighbouring spatial relationships.

Additional groundwater measurements would also provide opportunities for refinement. While arsenic concentration and depth provide essential information, other hydrogeological properties could further improve prediction capability. Variables such as aquifer characteristics, sediment composition, groundwater age, and additional chemical measurements may provide greater insight into the mechanisms controlling arsenic mobilisation.

Temporal information represents another important area for future data development. The current model primarily represents groundwater conditions using available historical measurements. However, arsenic concentrations can vary over time due to changes in groundwater movement, extraction patterns, and environmental conditions. The inclusion of repeated measurements from

the same wells could allow future models to learn temporal trends rather than relying solely on static spatial relationships.

As environmental datasets continue to improve, future versions of the model may therefore benefit from increased spatial resolution, improved chemical measurements, and long-term groundwater monitoring data.

6.5 Further Work

Several extensions could be explored to further develop the proposed approach.

One potential improvement is integrating a PointNet-based architecture alongside the existing CNN representation. The current model converts groundwater observations into a structured voxel grid, which enables convolutional processing but may introduce sparsity where observations are limited. A PointNet component could process irregular well locations directly, allowing the model to retain information from individual observations before combining this information with the voxel-based environmental representation.

Another area for future investigation is spatial validation. While the current evaluation demonstrates strong performance using a random split, a spatially separated test strategy would provide a stricter assessment of generalisation. This would involve training the model using wells from certain geographical regions and evaluating performance on unseen areas. Such testing would better represent deployment scenarios where predictions are required in locations with limited existing measurements.

6.6 Summary

This chapter evaluated the final 3D CNN model through confusion matrix analysis, benchmark comparison, and consideration of future improvements. The results demonstrate that the proposed approach achieved improved predictive performance compared with the existing iArsenic benchmark, supporting the potential of three-dimensional spatial deep learning for groundwater arsenic risk prediction.

Despite these improvements, further research is required to assess broader generalisation, incorporate additional environmental information, and investigate alternative model architectures.

Future developments should focus on improving data quality, incorporating temporal information, and increasing model transparency to support practical application within real-world arsenic monitoring systems.

Chapter 7

Reflection & Conclusion

7.1 Proposed Usage of Solution

The primary objective of this project was to investigate whether a three-dimensional convolutional neural network could provide an improved approach for groundwater arsenic risk prediction compared with existing methods. The final model demonstrated promising results, achieving substantial improvements over the selected iArsenic benchmark.

However, the practical value of this work extends beyond numerical performance metrics, as any proposed application must consider reliability, accessibility, and the limitations associated with predictive models in public health contexts.

The developed model has potential to support existing arsenic monitoring approaches by providing additional risk estimates for areas where laboratory testing is unavailable or impractical. Rather than replacing direct chemical analysis, the system could function as a prioritisation tool, identifying regions where further investigation or testing would provide the greatest benefit. This could allow limited resources to be directed towards communities where the likelihood of unsafe groundwater is highest.

Following completion of this project, the results and implementation will be shared with the iArsenic research team for further evaluation. This would allow domain experts to assess whether the additional spatial information provided by the 3D CNN approach offers practical advantages over existing models. Feedback from experts in hydrogeology and arsenic contamination would also be valuable in identifying the further environmental variables, validation methods, or model modifications that would best improve future versions.

A potential future application would be integrating the trained model into an accessible public-facing platform similar to existing iArsenic tools. However, deployment would require additional testing before being used in real-world decision-making. In particular, geographically separated validation would be necessary to confirm that the model can generalise effectively to areas where environmental conditions differ from those represented within the training dataset.

7.2 Research Questions Reflection

The research questions established at the beginning of this project provided the foundation for evaluating whether three-dimensional deep learning methods could contribute meaningful improvements to arsenic risk prediction. The completed development and evaluation process provides evidence towards answering both questions.

7.21 Research Question 1

“Is a 3D Convolutional Neural Network (CNN) a more viable approach for arsenic detection?”

The results of this project indicate that a 3D CNN is a viable and potentially more effective approach for arsenic risk prediction when sufficient spatial and environmental data are available. The final model achieved improved performance compared with the selected iArsenic benchmark across all evaluated metrics.

The primary advantage of the proposed approach is its ability to learn relationships between groundwater observations, environmental conditions, and depth information simultaneously.

Traditional approaches often represent wells as independent observations using limited characteristics such as depth and location. However, arsenic mobilisation is influenced by complex interactions involving geology, groundwater movement, and sediment conditions.

By representing the environment as a three-dimensional voxel structure, the model was able to incorporate both horizontal and vertical spatial relationships. The improvements introduced during development, particularly retaining depth information and adding surrounding risk distribution features, further demonstrated the importance of representing the subsurface environment appropriately.

However, these results do not prove that 3D CNNs are universally superior to all existing methods. Performance remains dependent on dataset quality, spatial resolution, and validation strategy. Further spatial testing is required to confirm generalisation outside the current validation approach.

Overall, the research question can be answered positively: a 3D CNN provides a promising approach for arsenic risk prediction by enabling models to capture spatial relationships that may not be represented through traditional feature-based methods.

7.22 Research Question 2

“What is the path forward for improving machine learning with chemical detection?”

The development of this project suggests that future improvements in machine learning for chemical detection will depend primarily on improving data quality, spatial representation, and interpretability rather than simply increasing model complexity.

One major limitation identified was the accuracy of available well locations. The current dataset relies on mouza-level information, meaning some wells are represented using approximate centroid locations. More precise coordinates would allow models to better capture local geological variation and improve the reliability of spatial relationships.

Future models could also benefit from additional hydrogeological and chemical variables. Arsenic mobilisation is controlled by complex interactions between geology, groundwater chemistry, sediment characteristics, and environmental conditions. Increasing the availability of these variables would allow machine learning models to better represent the underlying processes.

Another important area for future research is model interpretability. Although deep learning models can achieve strong predictive performance, understanding why predictions are generated is essential for public health applications. Explainable AI techniques could help identify the environmental factors contributing most strongly to predictions.

Future approaches could also combine multiple architectures, such as integrating voxel-based CNN models with point-based methods capable of processing irregular well locations directly. This may allow models to benefit from both structured environmental information and individual groundwater measurements.

7.3 Risks

Several risks identified during the project initiation stage influenced development. The most significant challenges involved data quality, computational requirements, and project scheduling.

Combining datasets from multiple sources introduced risks relating to spatial alignment and consistency. These were mitigated through preprocessing, coordinate transformation, and visual verification using QGIS before data was incorporated into the model.

Computational requirements were also a consideration due to the increased complexity of three-dimensional deep learning. This was managed through careful selection of voxel dimensions, patch sizes, and model architecture to balance performance with available hardware limitations. However, training was still slow overall and further enhancements could be made.

The project timeline also presented challenges, particularly because dataset preparation and debugging required more time than initially anticipated. However, the iterative development approach allowed priorities to be adjusted throughout the project, ensuring that meaningful progress could continue.

Overall, while several risks affected development, the mitigation strategies were effective in allowing the project to be completed successfully.

7.4 Project Management

The project followed an Agile-inspired iterative development approach, allowing changes to the dataset, architecture, and methodology based on experimental results. This was particularly appropriate for machine learning development, where successful approaches cannot always be predicted before implementation.

The original Gantt chart provided a useful structure for organising research, implementation, evaluation, and writing stages. However, the practical development process required greater flexibility, with some stages overlapping due to unexpected challenges during preprocessing and model development.

Version control through GitHub was valuable throughout development, allowing changes to be tracked and improving reproducibility. Maintaining separate preprocessing, training, and evaluation components also improved organisation and reduced the risk of errors.

A limitation of the project management approach was that insufficient time was initially allocated for final documentation and refinement.

Future projects of similar complexity would benefit from dedicating additional time towards analysis, visualisation, and report preparation.

7.5 Conclusion

This project investigated whether a three-dimensional convolutional neural network could improve groundwater arsenic prediction through the integration of spatial, environmental, and depth-dependent information.

The results demonstrate that the proposed approach achieved its primary objective, with the final model providing improved predictive performance compared with the selected benchmark. This supports the hypothesis that representing groundwater conditions as a three-dimensional environment allows machine learning models to capture relationships that may be missed by simpler approaches.

The project also demonstrated that effective environmental machine learning depends not only on model architecture but on appropriate data representation and domain understanding. The inclusion of spatial context, hydrogeological information, and depth relationships was central to the success of the approach.

Although further validation and refinement are required before practical deployment, this work demonstrates the potential of spatial deep learning as a supporting tool for arsenic risk assessment. Future research should focus on improving spatial accuracy, incorporating additional environmental information, validating performance across unseen regions, and increasing model interpretability.

Ultimately, this project provides evidence that advanced machine learning techniques can contribute towards improving environmental monitoring methods and supporting efforts to reduce the impact of groundwater arsenic contamination.

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AI Usage

Generative AI tools were used to assist with proofreading and improving the clarity of written sections. All content, analysis, and conclusions were independently produced and verified by the author.

Appendices

Appendix A

Project Initiation Document



Project Initiation Document

Project Details:

Project Title	<i>3D Spatial Deep Learning for Arsenic Risk Prediction</i>
Student	<i>Sam Derricott</i>
Course	<i>Software Engineering</i>
Project code	<i>PJE40</i>
Client	Bangladesh Public / I Arsenic Project
Supervisor	<i>John Cox</i>
Date	<i>October 2025</i>

Client / Target Audience

The target audience for my project is the people of Bangladesh affected by groundwater pollution as well as the iArsenic team.

Project background and problem to be solved

Groundwater arsenic contamination remains a significant public health issue in Bangladesh, where millions of people rely on groundwater from tubewells for drinking and domestic purposes. Existing prediction methods, such as those developed by the iArsenic project, provide valuable risk estimates using available well information; however, these approaches primarily rely on individual well characteristics and may not fully represent the complex environmental and geological processes influencing arsenic mobilisation.

The objective of this project is to investigate whether three-dimensional deep learning methods can improve arsenic risk prediction by incorporating groundwater measurements, environmental raster data, and spatial relationships into a voxel-based representation. The existing iArsenic model provides a benchmark accuracy of approximately 83.6%, which this project aims to improve through enhanced spatial feature representation.

Project aim and objectives

The aim of this project is to investigate whether a three-dimensional convolutional neural network (3D CNN) can improve groundwater arsenic risk prediction by learning spatial and environmental relationships within Bangladesh.

The objectives are:

- To preprocess and combine groundwater measurements with environmental geospatial datasets.
- To develop a three-dimensional voxel representation suitable for CNN-based spatial learning.
- To implement and evaluate a 3D CNN model for arsenic risk classification and concentration prediction.
- To compare model performance against existing iArsenic approaches using consistent evaluation metrics.
- To investigate how environmental features, spatial statistics, and architectural changes influence predictive performance.
- To analyse the limitations and potential future applications of deep learning methods for arsenic risk assessment.

Project constraints

- **Project deadline**

A fixed academic deadline limits the time available for research, implementation, testing, and evaluation.

- **Computational resources**

Deep learning models require significant computational resources. Training complexity must be balanced against available hardware capabilities.

- **Data availability and quality**

The model depends on the availability, accuracy, and spatial resolution of groundwater measurements and environmental datasets. Missing, inconsistent, or unevenly distributed data may influence prediction performance.

- **Model interpretability**

Deep learning models can be difficult to interpret compared with traditional statistical approaches. Additional analysis is required to justify predictions and understand feature importance.

- **Geographical validation**

The model is evaluated using available datasets from Bangladesh; therefore, performance outside the study region cannot be guaranteed.

Project management

An Agile iterative development approach will be used throughout the project. Development will be divided into stages consisting of research, dataset preparation, model implementation, evaluation, and refinement.

Each model iteration will be evaluated before further modifications are introduced. This allows unsuccessful approaches to be identified and ensures that architectural and feature changes are supported by measurable performance improvements.

Progress will be monitored through regular task planning, version control using GitHub, and meetings with the project supervisor.

Tasks and timescales

No	Stage	Dates	Main Tasks
1	<i>Project Initialisation & Research</i>	<i>October 2025 - November 2025</i>	- <i>Define project scope and objectives</i> - <i>Research groundwater arsenic contamination and existing prediction methods</i> - <i>Investigate the iArsenic project and establish project requirements</i>

			- Complete project initiation documentation and obtain ethics approval.
2	<i>Literature Review & Methodology Design</i>	<i>November 2025 - December 2025</i>	- Review existing machine learning and geospatial modelling approaches - Investigate CNN-based environmental prediction methods - Justify selected methodology, evaluation metrics, and technical approach - Identify required datasets and project resources.
3	<i>Data acquisition and pre-processing pipeline construction</i>	<i>December 2025 - January 2026</i>	- Obtain groundwater arsenic datasets - Clean and validate well measurements - Process environmental raster datasets - Generate voxel-based spatial representations - Prepare input data for model training.
4	<i>Initial model implementation</i>	<i>January 2026 - February 2026</i>	- Develop the initial 3D CNN architecture - Implement data loading and training pipelines - Establish baseline model performance

			- <i>Evaluate initial results against existing iArsenic benchmarks.</i>
5	<i>Continued development and refinement</i>	<i>February 2026 - March 2026</i>	- <i>Experiment with additional environmental input features</i> - <i>Modify model architecture based on evaluation results</i> - <i>Introduce regression alongside</i>
			<i>classification</i> - <i>Improve training stability and compare model iterations.</i>
6	<i>Evaluation and analysis</i>	<i>March 2026 - April 2026</i>	- <i>Analyse model performance using evaluation metrics</i> - <i>Compare final results against iArsenic</i> - <i>Analyse confusion matrices and prediction errors</i> - <i>Generate figures, tables, and results for dissertation.</i>

7	<i>Write-up finalisation</i>	<i>April 2026 - May 2026</i>	- <i>Complete dissertation chapters. - Refine evaluation, discussion, and conclusion sections</i> - <i>Finalise references and appendices</i> - <i>Update GitHub repository</i> - <i>Complete final proofreading and submission preparation.</i>
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Facilities and resources

- **Personal laptop** - for remote work - always available
- **Personal computer** - powerful enough to perform training efficiently enough - always available
- **University library** - remote work, study + resources around my chosen topic - available in-person or online if needed
- **Programming language** - Python, plus any documentation if needed
- **Software & libraries** - PyTorch, pandas, NumPy, Rasterio, plus any documentation or tutorials of such libraries if needed

Project risks

No	Description	Likelihood (high, medium, low)	Impact	Mitigation/Avoidance
1	Not saving after making changes	Medium	Loss of progress	Saving / committing regularly
2	Project being worked on on multiple devices	High	Poor management of project	Using GitHub
3	Poor time management	Medium	Incomplete or unfinished sections of project	Following the project management plan set out

4	Software / hardware failure	Medium	Loss of work / delays	Regular commits & save work online (GitHub)
5	Referencing / citation errors	Medium	Plagiarism accusations	Use of reference management software (e.g. Zotero, Mendeley)

Project deliverables

- Main dissertation document detailing the development, evaluation, and conclusions of the proposed 3D CNN arsenic risk prediction model
- A complete machine learning implementation including preprocessing pipelines, voxel generation, model architecture, training scripts, and evaluation code
- Visualisations of the dataset and model performance, including arsenic distribution analysis, spatial patterns, training progress, confusion matrices, and comparison against existing approaches
- A final comparison report analysing all model iterations, explaining architectural changes, feature improvements, performance differences, and conclusions regarding the effectiveness of 3D spatial deep learning
- Processed datasets, generated spatial features, and supporting files required to reproduce the experiments
- Analysis of patterns identified by the model, including relationships between environmental factors, groundwater characteristics, and arsenic risk
- Updated GitHub repository containing the final implementation and documentation

Research

- **Primary research:**
 - No primary research involving human participants is planned
 - Communication with members of the iArsenic research team may be conducted to gain additional technical or domain knowledge regarding existing arsenic prediction approaches
- **Secondary research:**
 - Academic literature relating to groundwater arsenic contamination, hydrogeology, machine learning, convolutional neural networks, geospatial modelling, and environmental prediction
 - Research papers evaluating artificial intelligence approaches for chemical and contaminant prediction
- Existing iArsenic publications, datasets, and technical documentation
 - Publicly available environmental datasets from recognised organisations, including geological, hydrological, and geographic information sources.

Legal, ethical, professional, social issues

- **Legal:**

- The project will use publicly available datasets and open-source software libraries. Appropriate attribution and referencing will be provided for all external datasets, research publications, and software dependencies
 - No personally identifiable information will be collected or processed
 - The project will comply with relevant university regulations regarding academic integrity, data usage, and research ethics
- **Ethical:**
 - The model is intended to support arsenic risk assessment and should not be considered a replacement for laboratory testing or professional water quality analysis, as such predictions will be presented as estimates of contamination risk rather than definitive statements about water safety
 - Care will be taken to reduce potential harm caused by incorrect predictions, particularly false negative classifications where unsafe wells may be incorrectly identified as safe
 - The limitations, uncertainty, and potential biases within the dataset and model will be clearly documented
 - Any use of artificial intelligence tools, including ChatGPT for proofreading or improving clarity, will be appropriately acknowledged in accordance with university guidelines
- **Professional:**
 - The project will follow professional software engineering practices, including version control, documentation, reproducible development processes, and structured testing
 - Model development decisions will be justified using experimental results and existing research rather than unsupported assumptions
 - Code quality will be maintained through organisation, documentation, and appropriate use of software development practices
 - Results will be reported objectively, including limitations and unsuccessful experiments, rather than only presenting successful outcomes
- **Social:**
 - The project addresses a significant public health challenge affecting communities in Bangladesh by investigating methods for improving arsenic risk prediction
 - Care will be taken to avoid overstating the capabilities of the model, as inaccurate predictions could influence decisions regarding drinking water safety
 - The intended outcome is to support improved identification of potential risk areas and provide additional information to existing monitoring approaches

Supervisor meetings

Meetings will be held regularly with supervisor John Cox to discuss progress and review development.

Declarations

Please tick the following declarations:

Usage (optional)	
☒	I give permission for this document to be made available to other students as examples of previous work.
Authenticity of work	
☒	I confirm that I have read the University rules in respect of plagiarism and student misconduct.
☒	I understand that if I use work from an external source, I must reference and cite the source in any work that I produce.
Ethics	
☒	I understand that ethical approval will be needed for this project regardless of whether I carry out primary research or not.
☒	I understand that if I do conduct primary research, I must include all raw data and research documentation in my final project report.
☒	I understand that not including the raw data can mean potentially failing the project.
☒	I understand that not obtaining ethical approval is grounds for academic misconduct and possible project failure.
Use of AI Artificial Intelligence (AI), such as ChatGPT, can be used as a tool to help assist and inform the initial development of your work but it is not a replacement for your own critical thinking and analysis. We require that your work is your own original content, demonstrating your knowledge, skills, and critical thinking abilities. To that end, AI-generated content must not be included in <u>any work</u> you submit for assessment unless suitably referenced. Not doing so is a breach of the University's Academic Regulations as outlined in the Student Conduct Policy and constitutes an assessment offence.	
☒	I have read the University AI policy and if relevant, I will acknowledge the use of AI in any work I produce for this project with appropriate referencing.

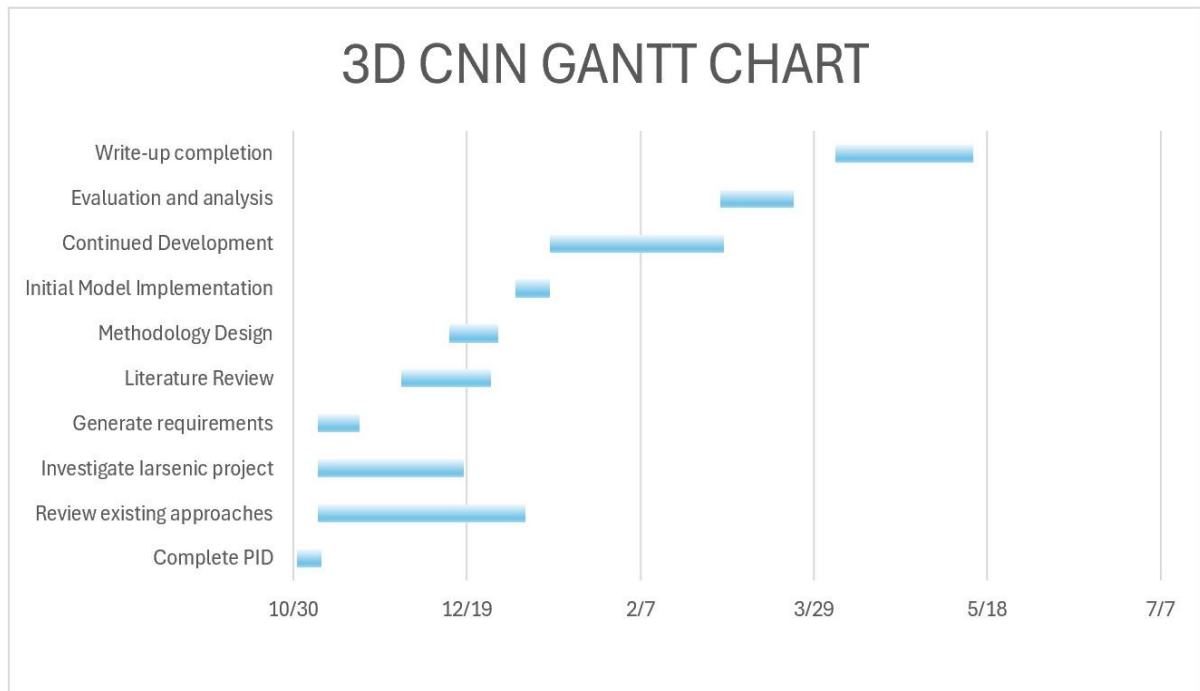


I understand that not appropriately referencing the use of AI is an assessment offence equal to plagiarism and may result in a penalty which remains on my student record.

Name: Sam Derricott

Date: 04/10/2

Gantt chart



Appendix B

Ethics Certificate



Certificate of Ethics Review

Project title: Enhancing Existing Predictive AI Models for Arsenic Contamination in Bangladeshi Water Sources

Name:	Sam Derricott	User ID:	up2247116	Application date:	19/11/2025 10:04:14	ER Number:	TETHIC-2025-112365
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You must download your referral certificate, print a copy and keep it as a record of this review.

The FEC representative(s) for the **School of Computing** is/are [Elisavet Andrikopoulou](#), [Kirsten Smith](#)

It is your responsibility to follow the University Code of Practice on Ethical Standards and any Department/School or professional guidelines in the conduct of your study including relevant guidelines regarding health and safety of researchers including the following:

- [University Policy](#)
- [Safety on Geological Fieldwork](#)

It is also your responsibility to follow University guidance on Data Protection Policy:

- [General guidance for all data protection issues](#)
- [University Data Protection Policy](#)

Which school/department do you belong to?: **School of Computing**

What is your primary role at the University?: **Undergraduate Student**

What is the name of the member of staff who is responsible for supervising your project?:
John Cox

Is the study likely to involve human subjects (observation) or participants?: No

Will financial inducements (other than reasonable expenses and compensation for time) be offered to participants?: No

Are there risks of significant damage to physical and/or ecological environmental features?: No Are there risks of significant damage to features of historical or cultural heritage (e.g. impacts of study techniques, taking of samples)?: No

Does the project involve animals in any way?: No

Could the research outputs potentially be harmful to third parties?: No

Could your research/artefact be adapted and be misused?: No

Will your project or project deliverables be relevant to defence, the military, police or other security organisations and/or in addition, could it be used by others to threaten UK security?: No

Please read and confirm that you agree with the following statements: I confirm that I have considered the implications for data collection and use, taking into consideration legal requirements (UK GDPR, Data Protection Act 2018 etc.), I confirm that I have considered the impact of this work and and taken any reasonable action to mitigate potential misuse of the project outputs, I confirm that I will act ethically and honestly throughout this project

Supervisor Review

As supervisor, I will ensure that this work will be conducted in an ethical manner in line with the University Ethics Policy.

Supervisor comments:

Supervisor's Digital Signature: **john.cox@port.ac.uk** Date: **27/02/2026**

Appendix C

Code Repository

The preprocessing pipelines, model architecture, training scripts, evaluation code, and associated data are provided in the GitHub repository. The repository contains the up-to-date implementation used throughout this project:

<https://github.com/SamD20/Arsenic-Risk-w-3D-CNN-PointNet->